Chiral quotients from one parity subgroup

We recover the quotient criterion and search of Hartley, Hubard and Leemans, *Two atlases of abstract chiral polytopes for small groups*, Ars Math. Contemp. 5 (2012), printed page 375. The underlying group-action argument is classical. Its role here is to specify one reusable interface, including the two different equivalence conventions in that paper. This note does not establish either of the paper's complete finite atlases.

Let $W = \langle r_0, \dots, r_{n-1} \rangle$ be a finite group generated by involutions, and $N \leq W$. On the right cosets Nw, put a colour-i edge from Nw to Nwr_i. Loops and coincident edges are permitted at this algebraic stage. The graph is connected because the r_i generate. When a quotient polytope is used below, we require its actual flag graph to be this graph. The source's semisparse quotient theorem supplies that geometric interface; an arbitrary orbit poset is not sufficient.

Theorem 1. *The colour-preserving automorphism group of the coset graph is $N_W(N)/N$. It acts freely, with $[W : N_W(N)]$ vertex orbits. If this is the flag graph of a polytope, that polytope is chiral if and only if $[W : N_W(N)] = 2$ and $r_i \notin N_W(N)$ for every i.*

Proof. A colour-preserving automorphism commutes with each of the permutations induced by r_i, hence with their products. If it sends N to Na, then it must send Nw to Naw. For $x \in N$, well-definedness at N requires $Nax = Na$, so $aNa^{-1} \leq N$. The two subgroups have the same finite order, hence equality. Thus $a \in N_W(N)$. Conversely any such a gives a well-defined bijection $Nw \mapsto Naw$ commuting with all colours. Two choices give the same map exactly when their cosets modulo N agree. This proves the asserted group identification.

If $Naw = Nw$, then $a \in N$, so the action is free. Its orbits therefore number

$$(|W|/|N|)/(|N_W(N)|/|N|) = [W : N_W(N)].$$

A polytope's automorphisms are precisely its colour-preserving flag-graph automorphisms: its i-faces are the components after deleting colour i, and face incidence is witnessed by flags. This uses the stipulated actual flag graph.

Write $H = N_W(N)$ and suppose $[W : H] = 2$, so $H \lhd W$. The vertices Nw and Nwr_i are in the same H/N-orbit exactly when $wr_iw^{-1} \in H$, equivalently $r_i \in H$. Consequently every adjacency exchanges the two orbits exactly when all r_i lie outside H. This is the defining flag-orbit condition for chirality. □

Proposition 2. *There is at most one subgroup H of index two avoiding every r_i. When it exists, the chiral quotients in Theorem 1 arise precisely from the subgroups $N \lhd H$ that are not normal in W and meet the geometric interface. Their conjugacy classes in W have two members, N and r_0Nr_0. The kernel of the flag action of W is*

$$C = \mathrm{Core}_W(N) = N \cap r_0Nr_0.$$

In H/C, the images of these two normal subgroups commute and intersect trivially. In particular, if the flag action is faithful their product is an internal direct product.

Proof. Every such index-two subgroup is the kernel of the homomorphism to C_2 sending each r_i to its nonidentity element. A map on a generating set determines a homomorphism uniquely. This also gives an effective existence test: assign parity to vertices of the Cayley graph, and

reject if any edge contradicts the assignment. Equivalently, the subgroup generated by the $r_0 r_i$ has index two; otherwise no such H exists. Indeed every even word is a product of pairs, and $r_i r_j = (r_0 r_i)^{-1}(r_0 r_j)$.

If $N \lhd H$ is not normal in W, its normalizer contains H but is not W, and therefore equals H. Conversely the normalizer condition of Theorem 1 implies exactly these requirements. Since $W = H \sqcup Hr_0$, conjugation produces just the two displayed subgroups. The usual transitive coset-action calculation gives its kernel as the intersection of all conjugates, hence the displayed C.

Both N and $r_0 N r_0$ are normal in H. The commutator of an element of one with an element of the other belongs to both, hence to C. Their images in H/C therefore commute and have trivial intersection; the product map from their direct product is an isomorphism onto its image. Finally, coset graphs of subgroups $N, M \leq W$ are colour-isomorphic exactly when N, M are conjugate: the same argument as above, for a map $N \mapsto Ma$, gives $aNa^{-1} = M$. Thus the two conjugates describe one unrooted quotient, rather than two different polytopes. □

The condition $C = 1$ is exactly faithfulness of this flag action. We do not infer that an arbitrary monodromy quotient is itself a regular polytope, or identify all notions of minimal polytopal cover. The source's qualified cover statement must be read with this distinction. Its small-cover algorithm removes the conjugate pair just proved. The almost-simple atlas instead reports both enantiomorphic markings before dividing by two; the acquired prefix-coordinate theorem treats those as oriented and unrooted equivalence, respectively.

A literal finite instance. The following instance exercises the interface without assuming any target atlas count. Over $\mathbf{F}_7$, put $V = \mathbf{F}_7^2$, $R = \mathrm{diag}(3, 5)$, and let S exchange the two coordinates. Then R has order six and $SRS = R^{-1}$. In $W = V \rtimes \langle R, S \rangle$, of order 588, use

$$r_0 = ((1, 1), R^3 S), \qquad r_1 = (0, RS), \qquad r_2 = (0, S).$$

These are involutions with pair orders $3, 6, 2$, in string order. They generate: $r_1 r_2 = R$, while $r_0 (R^3 S)^{-1}$ is translation by $(1, 1)$; this vector and its image $(3, 5)$ under R span V. The parity subgroup is $H = V \rtimes \langle R \rangle$.

The only normal subgroups of H that are not normal in W are the two coordinate lines. To see this, if a normal subgroup contains an element with rotation part $R^k \neq 1$, its commutators with V contain $(R^k - I)V = V$, because both diagonal entries of $R^k - I$ are nonzero. It is then the full inverse image of a subgroup of $H/V = C_6$, and is normal in W, since S acts on that cyclic quotient by inversion. Otherwise it is an R-invariant additive subgroup of V. Such a subgroup is an $\mathbf{F}_7$-subspace. The projections $(R - 5I)/(3 - 5)$ and $(R - 3I)/(5 - 3)$ show that it is a sum of coordinate lines. Of the four possibilities, only the two lines are not S-invariant.

The literal checker reconstructs all 588 group elements, verifies the full intersection property of the displayed regular marking, and builds the complete 84-flag quotient for either line. It checks the entire face poset, its flags, diamond intervals and every flag residue, not just the orders of adjacent products. Independently propagating every possible image of one vertex through the coloured graph gives exactly 42 automorphisms, agreeing with H/N. The lines are exchanged by r_0, intersect trivially, and yield one unrooted chiral quotient of this particular regular cover. Its mirror pair is not counted twice. This is a bounded validation example, not a proof of the source's complete $48 + 8$ or 17114 enumerations.